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ORIGINAL PAPER

Data-driven dynamic response forecasting and anomaly detection in long-span bridges

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Abstract

Long-span bridges, due to their high flexibility and low damping capacity, are particularly vulnerable to hazardous vibrations, such as vortex-induced vibrations (VIVs). Effective monitoring and early warning systems require advanced predictive models capable of accurately forecasting dynamic responses and detecting anomalies. This study proposes a Bayesian-optimized bidirectional long short-term memory sequence-to-sequence regression model for predicting vibrational responses and detecting abnormal structural behavior in long-span bridges. The proposed framework is designed with two configurations: one optimized for accurate prediction under normal conditions and another tailored to capture abrupt changes for anomaly detection. The model is trained and validated using structural health monitoring and weigh-in-motion data from an operational bridge, incorporating key environmental and operational conditions. By leveraging Bayesian optimization for hyperparameter tuning, the framework improves predictive accuracy and generalization. The model's performance is evaluated based on the prediction accuracy, the impact of time lag selection, and sensitivity of key input variables. Additionally, anomaly detection is achieved by analyzing residuals between predicted and measured responses, enabling early identification of hazardous events such as VIVs. The validation results demonstrate high predictive accuracy, with RMSE values below 0.05 m/s^2 and R2 exceeding 95%, alongside robust anomaly detection capabilities. These findings highlight the effectiveness of the proposed framework in enhancing the safety, reliability, and early warning capabilities of long-span bridges.

Keywords Bidirectional LSTM (Bi-LSTM) · Bayesian optimization · Structural health monitoring (SHM) · Anomaly detection · Vortexinducedvibration (VIV) · Long-span bridges

1 Introduction

Digital twin technology has been widely applied in structural health monitoring (SHM) to assess structural conditions and predict responses. Digital twins integrate real-time sensor data with computational models, allowing continuous evaluation of structural behavior and facilitating maintenance decisions [1, 2]. These models improve predictive accuracy by reducing uncertainties and enhancing anomaly detection, which is critical for maintaining long-span bridges under variable environmental and operational conditions [3, 4]. The increasing complexity of long-span bridges and their exposure to dynamic environmental loads have led to the adoption of digital modeling approaches for real-time structural assessment and early warning systems.

Structural response prediction relies on two primary modeling approaches: model-based and data-driven methods. The model-based approaches, such as finite element models (FEM), represent structural behavior using physics-based formulations and update analytical models based on measurement data to reduce discrepancies between predicted and observed responses [5]. However, these methods face challenges in capturing the complex nonlinear behavior of large-scale structures under changing environmental loads. For instance, vibration-based damage identification techniques have shown sensitivity to environmental variations, limiting their practical robustness [6]. Similarly, FEM-based approaches often struggle to incorporate temperature-induced deformations, introducing significant uncertainties in SHM application [4].

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To overcome these limitations, data-driven models have been increasingly explored. These methods utilize long-term multivariate SHM datasets to identify direct correlations between environmental and operational conditions (EOCs) and structural responses without requiring explicit physical modeling assumptions [7]. Machine learning techniques have proven effective in extracting complex patterns from large datasets, making them particularly suitable for SHM applications. Unlike model-based approaches, which require predefined structural parameters and aerodynamic force estimations, machine learning models autonomously learn relationships between EOCs and structural responses, improving predictive accuracy and adaptability to changing conditions.

Although data-driven models have been widely applied in structural response estimation, wind-induced vibrations present additional challenges for long-span bridges. These structures are particularly susceptible to vortex-induced vibration (VIV) and buffeting vibration due to their flexibility and low damping capacity [8–10]. Wind-induced excitations have been observed in multiple long-span bridges, raising concerns about fatigue damage and long-term performance degradation [9]. Predicting these responses is crucial for ensuring bridge safety and serviceability, especially since direct measurement of applied aerodynamic loads is infeasible.

Machine learning models, such as artificial neural networks (ANNs) and support vector regression (SVR), have been applied to wind-induced vibration prediction. ANNs and SVR have been successfully utilized to model the relationship between real-time EOCs and instantaneous vibrational amplitudes, particularly in predicting buffeting-induced responses [7]. Bayesian statistical models have also been employed to estimate lateral displacements of long-span bridges under varying wind conditions [11]. While these methods enhance real-time structural response estimation, their ability to forecast future bridge vibrations remains limited.

Forecasting vibrational responses over future time intervals is critical for early warning systems and proactive mitigation of hazardous conditions. Several studies have attempted to address this gap using machine learning frameworks, such as Gaussian process models and deep neural networks, to predict wind-induced bridge vibrations at future time steps [12]. Additionally, long short-term memory (LSTM) networks have been utilized to capture complex temporal dependencies for predicting structural responses under varying environmental conditions, including wind loads and temperature variations [13, 14]. However, standard LSTM models process time-series data unidirectionally, limiting their ability to fully leverage bidirectional dependencies in structural response forecasting.

In this regard, this study proposes a Bayesian-optimized bidirectional long short-term memory (Bi-LSTM) network for structural response forecasting and anomaly detection in long-span bridges. The Bi-LSTM model is designed to capture complex temporal dependencies inherent in dynamic structural responses. Unlike standard LSTM and recurrent neural network (RNN) models, Bi-LSTM processes time-series data in both forward and backward directions, improving the model's ability to capture long-range dependencies [15]. Furthermore, Bayesian optimization has been introduced as an effective method for hyperparameter tuning, improving model accuracy while reducing computational costs. By employing sequence-to-sequence regression architecture, the proposed framework enables robust predictions across varying time lags. Real-world SHM and weigh-in-motion (WIM) data collected from an operational long-span bridge are used to construct the training and testing datasets, ensuring the model's practical applicability. The study systematically integrates environmental and operational variables, such as temperature, wind speed, and traffic loads, to enhance the predictive performance of the Bi-LSTM model. The impact of a newly designed loss function and its key parameters is systematically examined to improve future prediction accuracy and further optimize model performance. Additionally, the trained model's capability for detecting unexpected vibrations, such as VIVs, is demonstrated using field monitoring datasets. These contributions represent a significant advancement, offering a scalable and adaptable methodology to enhance the reliability and safety of long-span bridges operating under dynamic conditions. The proposed framework is validated through comparative analyses with conventional LSTM and ARX models, demonstrating the advantages of bidirectional sequence learning and Bayesian Optimization in improving forecasting accuracy and early anomaly detection.

This paper is structured as follows: Sect. 2 details the data partitioning strategies and theoretical background of the Bi-LSTM model. Section 3 describes the target bridge and datasets used in the study. Section 4 presents the forecasting model's training and prediction performance, with a focus on hyperparameter optimization, time lag effects, and the role of EOCs. Section 5 explores anomaly detection methodologies using forecasting residuals, focusing on identifying VIVs. Finally, Sect. 6 concludes with a summary of the study's contributions, limitations, and future research directions.

2 Methodology

This section presents the methodology for predicting the root-mean-squared (RMS) amplitude of dynamic acceleration for the bridge deck after a certain time lag using the available SHM datasets. Figure 1 illustrates the schematic diagram of the proposed framework. First, the SHM dataset is divided into input and output data having a user-specified time lag. Subsequently, the stacked LSTM/Bi-LSTM model is trained on these training data, and the optimal

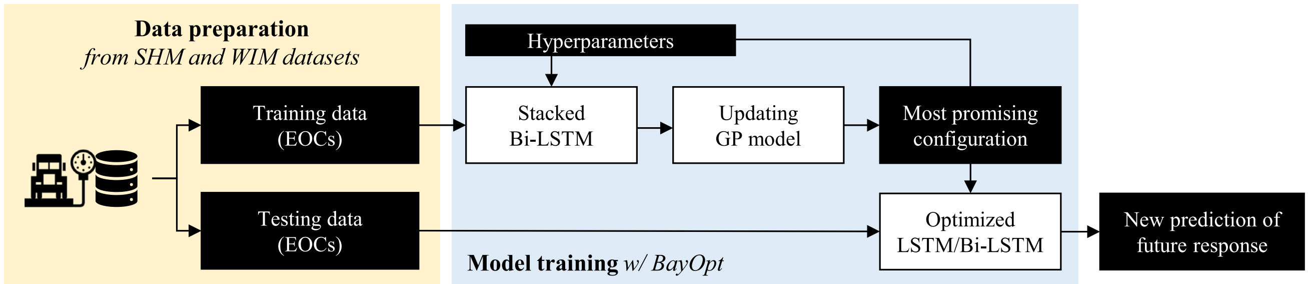


Fig. 1 Schematic diagram of the prediction model

hyperparameters are determined using Bayesian Optimization. The training datasets are composed of EOCs and the measured RMS acceleration values, and GP refers to the Gaussian process used for Bayesian optimization.

2.1 Data preparation for open loop forecasting

Forecasting approaches can be broadly categorized into two types: open-loop and closed-loop forecasting. Open-loop forecasting predicts the next step in a sequence using true input values from the data source, whereas closed-loop forecasting generates subsequent predictions by using previously forecasted values as inputs. Open-loop forecasting relies on continuous access to actual values to generate predictions for future time steps, while closed-loop forecasting operates independently of such real-time input.

In this study, open-loop forecasting was employed because true SHM and WIM data are continuously collected. To prepare the training data, representative statistics of the collected SHM dataset were calculated based on the desired unit of time. The mean value over each 10-min period was computed for features such as wind speed and temperature, while the RMS value was calculated for acceleration data to effectively represent its dynamic amplitude. For vehicle-related data, such as the number of vehicles and gross vehicle weight, cumulative values over the 10-min interval were used.

The prepared dataset was then partitioned into an input data, $\mathbf{X}(k)$, representing the k-th time step of the historical EOCs, and output data, $\mathbf{H}(k)$, representing the target values that the model aims to predict including the RMS acceleration values at specific future time steps. The input matrix of the model for T_{seq} fixed historical is established as follows:

$$\mathbf{X}(k) \in \mathbb{R}_{N\times T_{seq}} = \begin{bmatrix} U(k) & U(k+1) & \cdots & U\left(k+T_{seq}-2\right) & U\left(k+T_{seq}-1\right) \\ T(k) & T(k+1) & \cdots & T\left(k+T_{seq}-2\right) & T\left(k+T_{seq}-1\right) \\ n(k) & n(k+1) & \cdots & n\left(k+T_{seq}-2\right) & n\left(k+T_{seq}-1\right) \\ w(k) & w(k+1) & \cdots & w\left(k+T_{seq}-2\right) & w\left(k+T_{seq}-1\right) \\ a(k) & a(k+1) & \cdots & a\left(k+T_{seq}-2\right) & a\left(k+T_{seq}-1\right) \end{bmatrix} \quad (1)$$

In Eq. (1), $U(k)$ is the k-th mean wind speed over a 10-min interval, $T(k)$ is the mean temperature, $n(k)$ is the number of vehicles, $w(k)$ is the gross vehicle weight, and $a(k)$ is the RMS vertical acceleration, which serves as an indicator of the structural vibrational response. Each column in $\mathbf{X}(k)$ represents the values of the respective features over T_{seq} time steps, forming a time-series input to the model.

The output data, $\mathbf{H}(k)$, represents the RMS acceleration values at the time-lagged (τ) sequence, including the future prediction targets. Specifically, $\mathbf{H}(k)$ consists of two parts: (1) a subset of the last row of $\mathbf{X}(k)$, which corresponds to the RMS acceleration values from $a(k+\tau)$ to $a(k+T_{seq}-1)$, and (2) the unknown future values from $a\left(k+T_{seq}\right)$ to $a\left(k+T_{seq}+\tau-1\right)$, which the model is trained to predict. This relationship is defined as follows:

$$\mathbf{H}(k) \in \mathbb{R}_{1\times T_{seq}} = \left[\, a(k+\tau) \;\; a(k+\tau+1) \;\; \cdots \;\; a\left(k+T_{seq}\right) \;\; \cdots \;\; a\left(k+T_{seq}+\tau-1\right) \,\right] \quad (2)$$

Note that the values from $a(k+\tau)$ to $a(k+T_{seq}-1)$ are already included in the input matrix and serve as part of the training data, while the final τ elements represent new future RMS accelerations to be forecasted. To define the prediction targets, we denote them as $\mathbf{P}(k)$, which corresponds to the final τ element of $\mathbf{H}(k)$, as follows:

$$\mathbf{P}(k) \in \mathbb{R}_{1\times\tau} = \left[\, a\left(k+T_{seq}\right) \;\; a\left(k+T_{seq}+1\right) \;\; \cdots \;\; a\left(k+T_{seq}+\tau-1\right) \,\right] \quad (3)$$

These values represent the expected RMS acceleration response at future time steps, given the historical environmental and operational conditions provided in $\mathbf{X}(k)$. This formulation allows the model to leverage past patterns in acceleration data while explicitly distinguishing between known values used during training and new predictions to be made.

Figure 2 illustrates how the 10-min statistical datasets are arranged. In this example, the total number of features is 5, the time step k is 1, the sequence length T_{seq} is 144 (representing one day's data), and the time lag is 20 min ($\tau = 2$). The green box highlights the RMS acceleration values at time steps 145 ($= k + T_{seq}$) to 146 ($= k + T_{seq} + \tau - 1$), representing the unknown future responses the model is trained to predict.

2.2 Stacked Bi-LSTM with Bayesian optimization

2.2.1 Stacked Bi-LSTM network

In this study, a stacked Bi-LSTM network architecture is used for the sequence-to-sequence prediction model. Unlike a standard LSTM network, the Bi-LSTM processes sequential data in both forward and backward directions, allowing it to capture contextual information from both past and future time steps. In a stacked Bi-LSTM architecture, multiple Bi-LSTM layers are stacked hierarchically, with each layer taking the output of the previous layer as input. Within each Bi-LSTM unit, forward and backward LSTM layers operate in opposite directions, and their outputs are concatenated to generate the final output for that layer. Stacking multiple Bi-LSTM layers potentially enhances the model's capacity to capture complex temporal dependencies in the input data, thereby improving accuracy in tasks such as sequence prediction and classification. However, while this increased model capacity can lead to better performance, it also elevates the risk of overfitting, particularly with limited training data. To address this, Bayesian optimization was employed to determine the optimal hyperparameter configuration for the stacked Bi-LSTM model, balancing accuracy and generalization.

Figure 3 presents a schematic diagram of the two-module stacked Bi-LSTM network, illustrating sequential data flow and layer connections. This corresponds to the example of data configuration shown in Fig. 2, where $k = 1$, $\tau = 2$, and $T_{seq} = 144$. Each LSTM cell (green box) follows a standard state update mechanism, where the cell state activation function is tanh, and the gate activation functions (input, forget, and output gates) are sigmoid functions. A single Bi-LSTM module, highlighted as gray box, consists of three layers: a Bi-LSTM layer, a fully connected (FC) layer, and a dropout layer to prevent overfitting. If two stacked Bi-LSTM modules are used, the layers are arranged sequentially in the order Bi-LSTM–FC–Dropout–Bi-LSTM–FC–Dropout. Finally, the network concludes with a FC layer, followed by a regression output layer for final predictions.

2.2.2 Bayesian optimization

The Bayesian optimization process aims to identify the optimal hyperparameters for the Bi-LSTM model by iteratively evaluating candidate configurations. The process begins with an initial evaluation, where a set of sample points is randomly selected within the defined hyperparameter search space. The objective function, representing model validation performance, is computed for these initial points. If an evaluation fails, additional random points are selected until successful evaluations are obtained.

A Gaussian process regression model is then constructed based on the observed data, estimating the posterior distribution of the objective function. Using this model, the optimization algorithm maximizes an acquisition function, which determines the next candidate point for evaluation by balancing exploration and exploitation. The acquisition function used in this study was expected improvement per second plus chosen to balance computational efficiency and accuracy in the hyperparameter search.

The optimization process continues iteratively, updating the Gaussian process model and selecting new hyperparameter configurations. The procedure terminates when a predefined stopping criterion is met, such as reaching the maximum number of iterations (set to 100 in this study) or achieving convergence. The final set of optimized hyperparameters is then applied to train the Bi-LSTM model, ensuring enhanced predictive accuracy and generalization.

08/01 00:00	08/01 00:10	08/01 00:20	08/01 00:30	08/01 00:40	08/01 00:50	08/01 01:00	…	08/01 23:50	08/01 24:00	08/01 00:10	08/01 00:20	…
$U(1)$	$U(2)$	$U(3)$	$U(4)$	$U(5)$	$U(6)$	$U(7)$	…	$U(143)$	$U(144)$	$U(145)$	$U(146)$	…
$T(1)$	$T(2)$	$T(3)$	$T(4)$	$T(5)$	$T(6)$	$T(7)$	…	$T(143)$	$T(144)$	$T(145)$	$T(146)$	…
$n(1)$	$n(2)$	$n(3)$	$n(4)$	$n(5)$	$n(6)$	$n(7)$	…	$n(143)$	$n(144)$	$n(145)$	$n(146)$	…
$w(1)$	$w(2)$	$w(3)$	$w(4)$	$w(5)$	$w(6)$	$w(7)$	…	$w(143)$	$w(144)$	$w(145)$	$w(146)$	…
$a(1)$	$a(2)$	$a(3)$	$a(4)$	$a(5)$	$a(6)$	$a(7)$	…	$a(143)$	$a(144)$	$a(145)$	$a(146)$	…

EOV: **X**(1) [5x144] Future RMS acceleration: **H**(1) [1x144] New Prediction: **P**(1) [2x1]

Fig. 2 The example of data preparation ($T_{seq} = 144$, $k = 1$, $\tau = 2$)

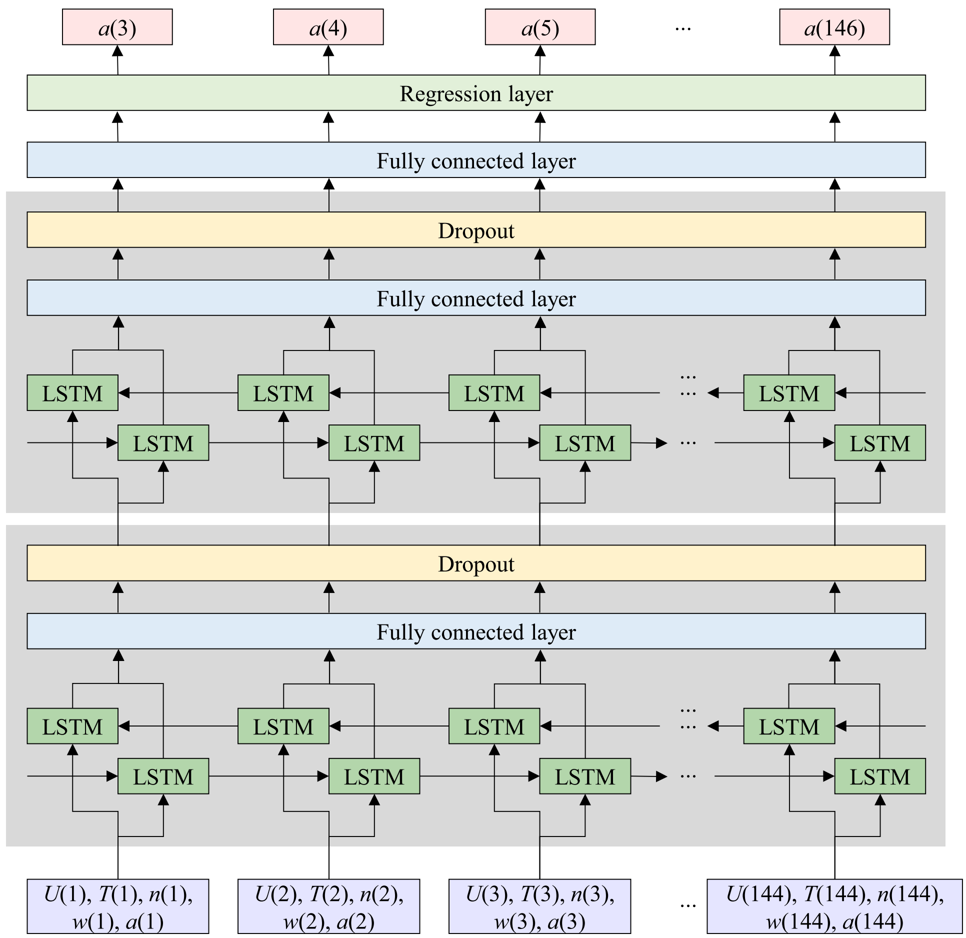


Fig. 3 Schematic of a two-module stacked Bi-LSTM network ($T_{seq} = 144$, $k = 1$, $\tau = 2$)

Table 1 Optimizable hyperparameters

Name	Interval	Transformation type
Initial learning rate	[8e-3, 1e-1]	Log
L2 regularization factor	[1e-5, 1e-2]	Log
The number of hidden units	[50, 200]	Integer
The number of stacked LSTM layers	[1, 4]	Integer
The usage of Bi-LSTM	[0, 1]	Logical

Table 1 summarizes the optimizable hyperparameters used in this study. The binary hyperparameters, such as logical switches, are represented with a value of 0 indicating false and 1 indicating true. This optimization process ensured that the model's complexity and performance were aligned with the dataset characteristics and the forecasting task.

The employed loss function for Bayesian optimization, $\mathcal{L}_{BO}$, is defined as a weighted linear combination of the average root-mean-square errors (RMSE) of the total sequential RMS acceleration ($\mathbf{H}$) and their future RMS acceleration part ($\mathbf{P}$), with a predefined weight α. This formulation enables a trade-off between short-term sequence learning and long-term forecasting accuracy and is expressed as:

$$\mathcal{L}_{BO} = (1 - \alpha)\frac{\sum_{k=1}^{T_{seq}} \mathrm{RMSE}\left(\mathbf{H}(k), \hat{\mathbf{H}}(k)\right)}{T_{seq}} + \alpha \frac{\sum_{k=1}^{\tau} \mathrm{RMSE}\left(\mathbf{P}(k), \hat{\mathbf{P}}(k)\right)}{\tau} \quad (4)$$

In Eq. (4), the RMSE is computed using the L2-norm operator to quantify the difference between measured and predicted values, where the hat notation (^) denotes model-predicted value.

Unlike conventional sequence-to-sequence forecasting methods, which only predict future steps, our model incorporates overlapping RMS acceleration sequences to improve both historical continuity and predictive learning. This design allows the model to leverage past data more effectively while ensuring smooth transitions in acceleration patterns. By optimizing over $\mathbf{H}$, the model learns to maintain consistency between past and future acceleration values, thereby reducing the likelihood of sharp discontinuities.

In addition, the inclusion of $\mathbf{P}$ in the loss function ensures that, while the model captures sequential dependencies from past sequences, it prioritizes accurate forecasting of future RMS accelerations. The weighting parameter α allows for flexible adjustment of the model's focus between preserving past sequence consistency and improving long-term prediction accuracy, depending on the specific application requirements. The use of overlapping sequences further reinforces the model's ability to capture gradual trends in acceleration patterns, mitigating abrupt shifts in predictions while maintaining stable transitions over time.

2.3 Performance indicators

The performance of the trained forecasting model was evaluated using two widely adopted metrics: RMSE and coefficient of determination (R2). RMSE quantifies the average magnitude of the difference between the forecasted and actual responses, with smaller RMSE values indicating more accurate predictions. R2, on the other hand, measures the proportion of variance in the dependent variable explained by the independent variables in the model. R2 values range from 0 to 1, where higher values signify a better fit of the model to the data. Together, these metrics provide a comprehensive evaluation of the model's predictive accuracy and its ability to explain observed variations in the data.

3 Data

3.1 Target structure

This study investigates the Jindo Bridge, a twin cable-stayed bridge system located in South Korea. The first bridge, completed in 1984, was later complemented by a second bridge, referred to as Bridge 2, in 2005 to accommodate increased traffic demands. Both bridges feature a two-lane configuration, with a main span of 344 m and two side spans of 73.79 m each, serving unidirectional roadways. The structures employ steel box girders and steel pylons, exhibiting comparable configurations, dimensions, cross-sectional profiles, and dynamic characteristics, as depicted in Fig. 4.

However, the addition of Bridge 2 introduced notable aerodynamic interactions between the two decks, which triggered significant VIV in Bridge 2 [16]. Further investigation identified the low damping capacity of Bridge 2 as a primary factor exacerbating these vibrations [8]. In response, a multiple-tuned mass damper (MTMD) was installed at the mid-span of Bridge 2 in 2014. This modification substantially enhanced the damping characteristics, increasing the fundamental mode damping ratio from 0.29 to 3.30% [17]. These structural improvements aimed to mitigate VIV and ensure the long-term operational performance of the bridge.

3.2 Monitoring data

Long-span bridges, such as the Jindo Bridge, are subject to ambient vibrations primarily induced by wind and traffic. These factors must be carefully monitored and analyzed as they play a critical role in predicting vibrational amplitudes and ensuring structural safety. To address this, the data were collected using SHM and weigh-in-motion (WIM) systems, which were strategically deployed to capture relevant environmental and operational features. The SHM system consists of built-in sensors installed at the center of the main span, including two accelerometers (Kinematics, EpiSensor ES-U2), an ultrasonic anemometer (Gill Instrument, WindMaster Pro), and multiple thermometers (GTC Corporation, NTC-3 KD-3 F). These sensors provided data after the installation of the MTMD, enabling precise measurement of vibration and environmental conditions. The sampling frequencies were set to 100 Hz for acceleration and temperature

Fig. 4 Jindo Bridge

measurements and 1 Hz for wind speed. Simultaneously, the WIM systems located at the bridge entrance recorded vehicle-related parameters such as vehicle type, gross vehicle weight (GVW), length, class, and crossing time. Traffic-related features, including the number of vehicles and total GVW, were extracted every 10 min from the WIM data. Table 2 lists the detailed specifications of the accelerometer, ultrasonic anemometer, and thermometers employed in the SHM system.

A 3-month monitoring dataset, spanning June 1 to September 6, 2019, was utilized for this study. Five key features were analyzed: RMS acceleration, mean wind speed, temperature, vehicle count, and total GVW. Figure 5 illustrates the temporal trends of these features over one representative week. Measured values are depicted by gray dots, while blue lines represent their moving averages. Both SHM and WIM data exhibited consistent daily patterns, with vibrational activity peaking during daytime hours and diminishing overnight, resembling a half-sine wave. Leveraging these patterns, input and output arrays were constructed at daily intervals, each comprising 144 data points, as depicted in Fig. 2. Additional details on the SHM/WIM dataset can be found in Kim and Kim [18].

During the monitoring period, VIV events were limited to a typhoon event, during which the MTMD was locked to prevent excessive vibration of the masses under strong wind conditions. To facilitate analysis, the SHM and WIM datasets were categorized into two groups: (1) D1: training datasets without VIV, and (2) D2: a one-week test dataset containing VIV events. The models in this study were trained and validated using D1, while anomaly detection, as detailed in Sect. 5, was performed on D2. Figure 6 presents the velocity–amplitude (V-A) curves for both datasets, along with their respective time-series descriptive statistics over the monitoring period. This separation of datasets ensures robust training and evaluation of the proposed methods under both typical and anomalous operational conditions.

4 Forecasting model

This section presents the experimental results derived from actual datasets, including evaluations of feature importance, weight coefficients, and time lag effects on the forecasting of vibrational responses. Additionally, the impact of including anomalies (e.g., VIV cases) in the training data is assessed, considering the challenges of obtaining sufficient

Table 2 Detailed specifications of the built-in sensors

Sensors	Parameters	Value
Accelerometer (EpiSensor ES-U2)	Type	Uniaxial force-balance
	Dynamic range	155dB+
	Bandwidth	DC to200 Hz
	Full-scale range	±0.25g to ±4g
	Linearity	< 1000µg/g2
	Hysteresis	< 0.1% of full scale
	Operating temperature	-20°C to 70°C
Anemometer (WindMaster Pro)	Type	Ultrasonic anemometer
	Range (speed)	0 – 65 m/s
	Resolution (speed)	0.01 m/s
	Accuracy (speed)	< 1.5% RMS @ 12 m/s
	Range (direction)	0 – 360° (direction)
	Resolution (direction)	0.1°
	Accuracy (direction)	2° @ 12 m/s
	Operating temperature	-40°C to 70°C
Thermometer (NTC-3KD-3F)	Type	Thermistor
	Temperature Range	-40°C to 120°C
	R@25°C	3.0 kΩ
	Beta (25/85)	3,560±1.0%

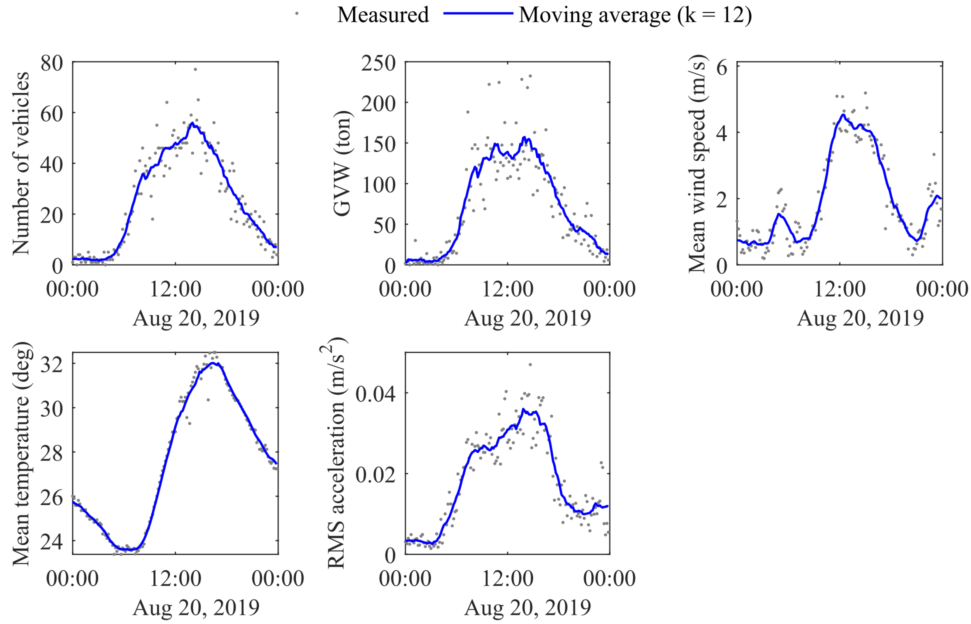


Fig. 5 Daily trend of EOCs on August 20, 2019

anomaly datasets in real-world applications. The dataset was partitioned using the 80:20 holdout rule to ensure an unbiased estimate of model performance on unseen data, while minimizing overfitting. Experiments were conducted on a Bi-LSTM network implemented using the MATLAB deep-learning platform on a system equipped with an Intel i5-11,500 2.70 GHz processor, 16 GB RAM, and a GeForce RTX 2060 GPU.

4.1 Training results with hyperparameter optimization

The results of Bayesian optimization for the weight coefficient $\alpha = 0.3$ are shown in Fig. 7. The x-axis represents the iteration number, while the y-axis corresponds to the minimum objective function value. The "observed" value in the legend indicates the minimum objective value across all observed points at each iteration, and the "estimated" value represents the estimated objective function value determined using the minimum upper confidence bound at the corresponding iteration. Convergence of the objective function was achieved rapidly, with values stabilizing before the 50 th iteration for all cases.

The optimal hyperparameters for different time lags (τ) are summarized in Table 3. All configurations consistently employed a single Bi-LSTM layer, underscoring its effectiveness for the given forecasting tasks. While slight variations were observed across time lags, the hyperparameters followed a generally consistent trend. For instance, the initial learning rate ranged between 0.01 and 0.02, and the L2 regularization factors were consistently within 1.00×10^{-5} to 2.00×10^{-5}. Similarly, the number of hidden units remained approximately between 180 and 190. These results confirm that the Bi-LSTM network is well-suited for predicting the dynamic responses of bridges, with a single-layer configuration proving sufficient for the task. Overall, the hyperparameter consistency across different time lags highlights the robustness and adaptability of the proposed framework.

4.2 Predictive performance according to time lags

The predictive performance of the proposed framework was evaluated by comparing the predicted vibrational responses with actual measurements across different time lag ($\alpha = 0.3$). Figure 8 presents the time histories of the true and predicted RMS amplitudes for a validation dataset derived from the 20% holdout portion of D1, representing normal operational conditions. The measured (true) RMS accelerations are shown as blue circles, while predicted values are represented by red x marks. Horizontal gray lines segment the data into individual days, highlighting daily fluctuations. The scatter plots further illustrate the agreement between predicted and true values.

The results demonstrate excellent predictive accuracy for time lags up to 30 min, where all models effectively predicted both low- and high-amplitude vibrational responses. However, a noticeable decline in performance was observed for time lags exceeding 40 min, with significant discrepancies in high-amplitude predictions becoming evident at the 60-min mark, as highlighted in the blue boxes. These trends are further confirmed by the RMSE and R2 metrics across different time lags, as shown in Fig. 9. Despite the degradation in accuracy at longer prediction horizons, the Bi-LSTM

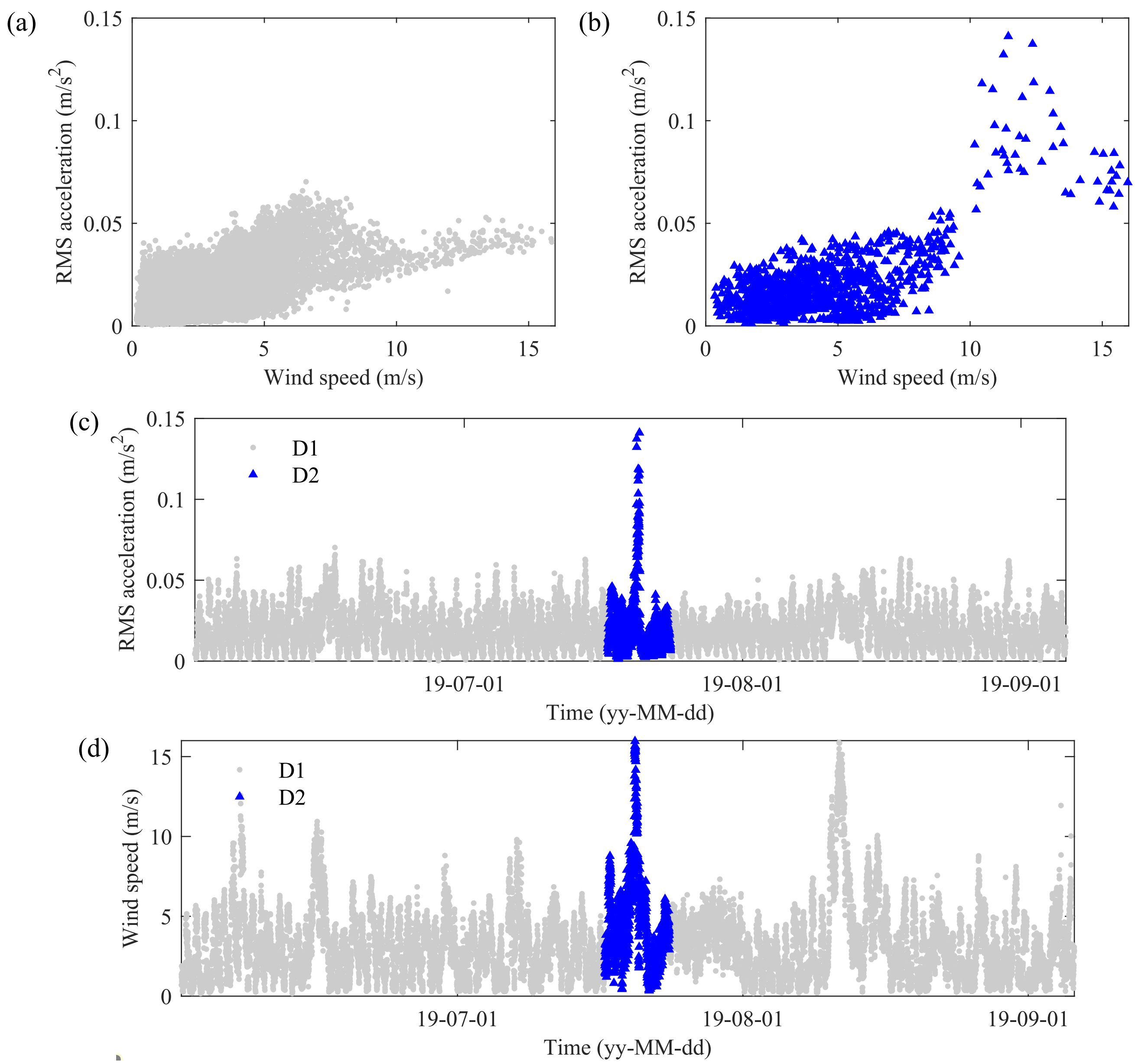


Fig. 6 Descriptive statistics of the monitoring data: (**a**) V–A curve of D1, (**b**) V–A curve of D2, and time series of (**c**) RMS acceleration and (**d**) mean wind speed

network maintains strong reliability for dynamic response forecasting with a lead time of 30 min.

The impact of α on the performance of the trained model are examined in Fig. 10. Generally, an increase in α, which represents the weight assigned to future predictions, generally leads to a reduction in prediction accuracy. This decline is primarily due to the composition of the loss function, where future-prediction elements are relatively sparse. As a result, the model tends to overfit by focusing disproportionately on a minor subset of the data, which destabilizes the loss function and impairs overall performance. Conversely, lower values of α allow the model to effectively capture the overall daily trends, which acts as a regularization mechanism, enhancing generalization capability.

4.3 Comparative analysis of the proposed Bi-LSTM with different models

To further validate the advantages of the proposed Bi-LSTM network and the impact of Bayesian optimization, additional comparative experiments were conducted. The primary objective of these comparisons is to confirm that the proposed approach not only enhances predictive accuracy through Bayesian optimization but also effectively captures

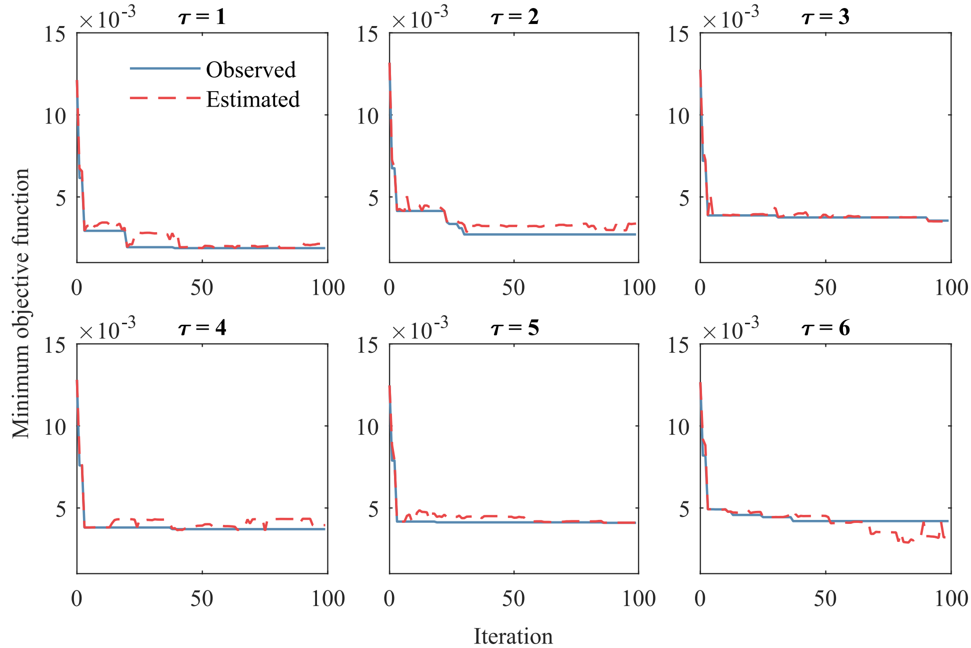


Fig. 7 Results of Bayesian optimization (α = 0.3)

Table 3 Optimal hyperparameters according to the time lag (α = 0.3)

Name	τ = 1	τ = 2	τ = 3	τ = 4	τ = 5	τ = 6
Initial learning rate	0.0158	0.0119	0.0179	0.0093	0.0082	0.0172
L2 regularization factor	1.04E-05	1.31E-05	1.67E-05	1.13E-05	2.16E-05	1.13E-05
The number of hidden units	185	194	191	163	195	189
The number of stacked LSTM layers	1	1	1	1	1	1
The usage of Bi-LSTM	1	1	1	1	1	1

structural response dynamics that conventional linear models and alternative LSTM-based architectures struggle to represent.

First, to evaluate the necessity of using a nonlinear Bi-LSTM network for predicting structural responses, its performance was compared against two widely used linear models: auto regressive with exogenous input (ARX) and generalized linear model (GLM). These models were selected as representative linear approaches, where ARX explicitly accounts for time-series dependencies, while GLM provides a flexible framework for modeling relationships between environmental and structural response variables. The input and output for both linear models are set to $\mathbf{X}(k)$ and $\mathbf{H}(k)$, respectively.

To further evaluate the effectiveness of bidirectional learning, a Bayesian-optimized LSTM was assessed by comparing its performance with the optimized Bi-LSTM model. In addition, the impact of Bayesian optimization was examined through different Bi-LSTM configurations, where the learning rate, L2 regularization coefficient, and number of hidden units were randomly selected within the Bayesian optimization search space. The evaluated configurations include: (1) Bi-LSTM (unoptimized): Bi-LSTM with randomly selected hyperparameters, (2) Bi-LSTM (± 50%): Bi-LSTM with hyperparameters randomly varied within 50–150% of the optimized values, and (3) Bi-LSTM (± 20%): Bi-LSTM with hyperparameters randomly varied within 80–120% of the optimized values. Bayesian optimization was conducted for 100 iterations per configuration, and the median R2 value across these iterations was selected as the performance metric to ensure robustness against extreme outliers.

The performance comparison is summarized in Table 4. The results demonstrate that while all LSTM-based models outperform linear models (GLM, ARX), Bi-LSTM achieves the highest accuracy, particularly when optimized with Bayesian tuning. The linear models of GLM and ARX yielded similar RMSE values of 0.0066 m/s^2, with corresponding R2 values of 52.27% and 50.81%, respectively. These results indicate that linear models, although capable of capturing broad trends in RMS acceleration, fail to model the complex nonlinear relationships between environmental variables and structural responses, leading to poor predictive performance Table 4.

Replacing linear models with an optimized unidirectional LSTM slightly improved prediction accuracy, reducing

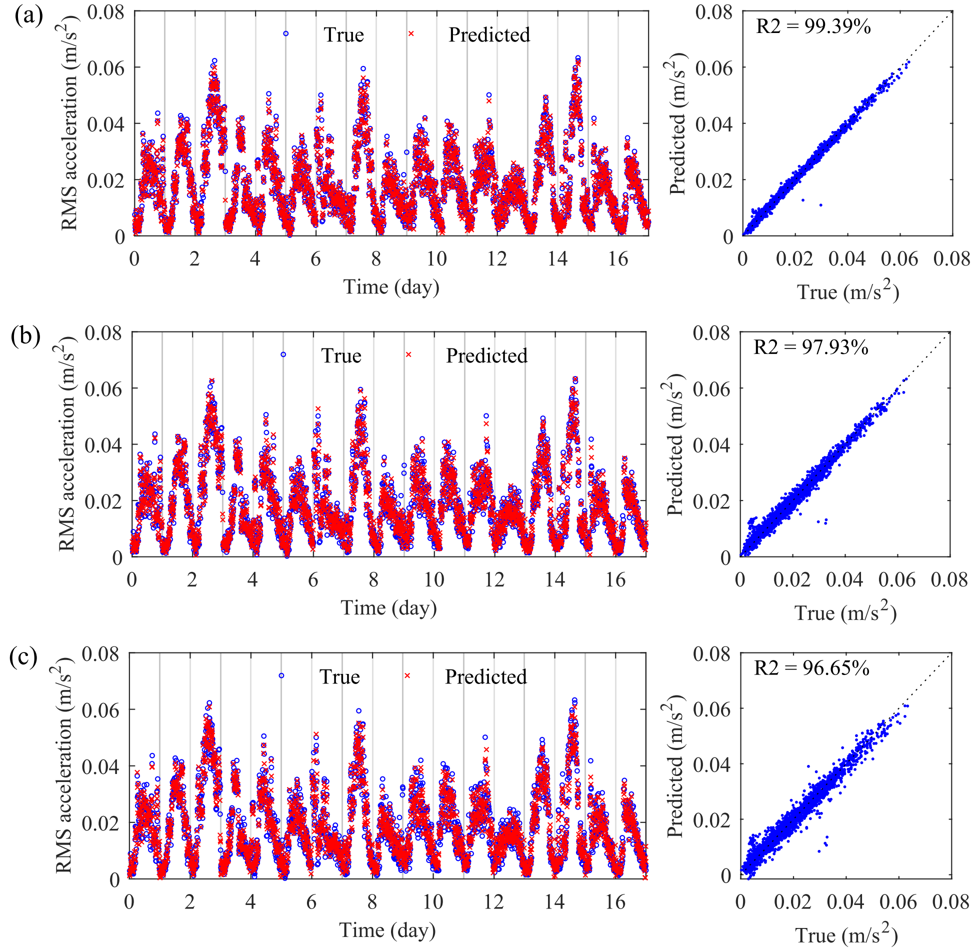


Fig. 8 The results of predicted vibrational response: (**a**) τ = 1, (**b**) τ = 2, (**c**)τ = 3, (**d**) τ = 4, (**e**) τ = 5, and (**f**) τ = 6

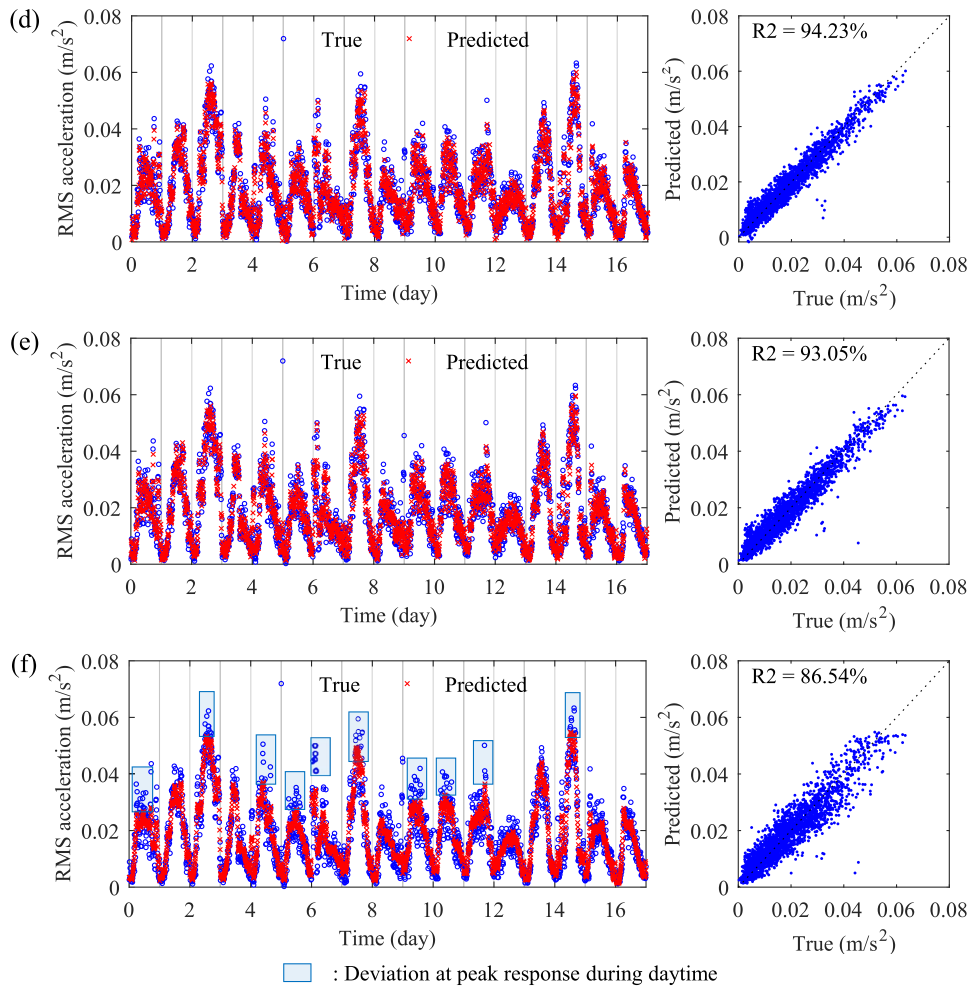
RMSE to 0.0062 m/s^2 and increasing R2 to 56.31%. However, when transitioning to a Bi-LSTM architecture, even without hyperparameter tuning, RMSE significantly decreased to 0.0046 m/s^2, while R2 improved to 75.68%, highlighting the advantage of bidirectional sequence learning in capturing structural response dependencies. Bayesian optimization further enhanced the model's predictive performance. The Bi-LSTM with randomly selected hyperparameters exhibited moderate performance, but the random selection within ± 50% of the optimal setting improved R2 to 91.95%, with a corresponding RMSE reduction to 0.0026 m/s^2. When the hyperparameter values were varied within ± 20% of the optimal setting, R2 further increased to 92.81%, approaching the accuracy of the fully optimized Bi-LSTM. These findings confirm that while Bi-LSTM inherently improves predictive performance over unidirectional LSTMs and linear models, its full potential is realized when paired with the optimal configuration of hyperparameters.

4.4 Feature importance and predictive performance without prior RMS acceleration

Feature engineering was performed to evaluate the significance of individual variables for vibrational response prediction, focusing on a 30-min lag (τ = 3). Figure 11 compares the model performance with the α of 0.3 when specific features were excluded from the training dataset. For this analysis, each feature was excluded individually to assess its impact on the model's predictive performance. For example, the exclusion of "*w*" (gross vehicle weight) indicates that the network was trained to predict future RMS acceleration using only temperature (*T*), mean wind speed (*U*), the number of vehicles (*n*), and RMS acceleration (*a*). The red dashed line indicates the RMSE and R2 values when none of the features are excluded.

The results confirm that all features contribute to the accuracy of predicting future dynamic responses, as their exclusion leads to increased RMSE and decreased R2 values. Among these features, RMS acceleration

Fig. 8 (continued)



was identified as the most critical variable. Its exclusion resulted in a significant drop in the R2 value from 96.65 to 64.27%, indicating a substantial loss in predictive accuracy. In contrast, the exclusion of other EOCs, such as mean wind speed, temperature, total GVW, and the number of vehicles, had relatively smaller impacts, with R2 values of 91.36%, 91.74%, 91.80%, and 94.49%, respectively. These findings suggest that prior models relied predominantly on the temporal dependency of RMS acceleration, which highlights its central role in dynamic response prediction.

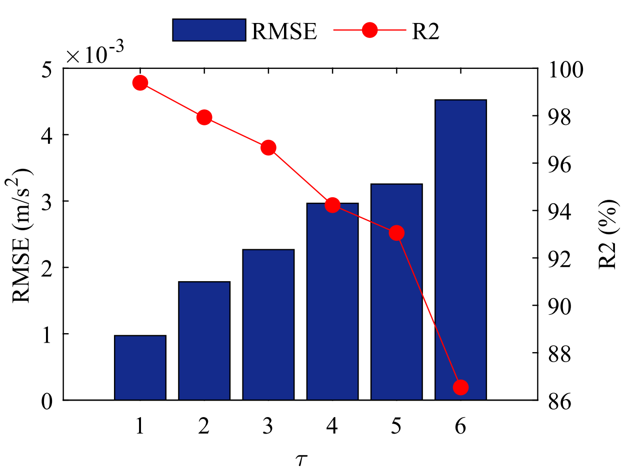


Fig. 9 Prediction performance according to the different time lag

To further explore this, a new Bi-LSTM model was trained using four features: mean wind speed, temperature, the number of vehicles, and total GVW, excluding prior RMS acceleration. Predictive performance was assessed for time lags of 10, 20, and 30 min with α = 0.3. As shown in Fig. 12, the results indicate that the model trained without prior RMS acceleration exhibited degraded performance compared to the models incorporating prior RMS acceleration, as discussed in Sect. 4.2. This suggests that prior RMS acceleration plays a significant role in enhancing the predictive accuracy of future dynamic responses. This degradation was particularly evident in high-amplitude regions, where discrepancies between predictions and ground truths were

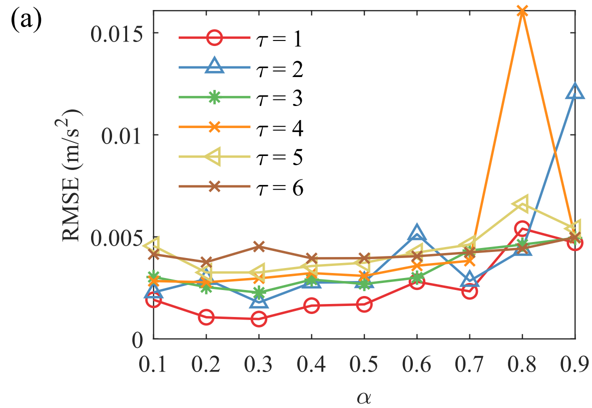


Fig. 10 Impact of α on the model performance: **a** RMSE and **b** R2

Table 4 Performance comparison of different model configurations (τ =3, α= 0.3)

Model	RMSE (m/s^2)	R2 (%)
GLM	0.0066	52.27
ARX	0.0066	50.81
LSTM (Optimized)	0.0062	56.31
Bi-LSTM (Unoptimized)	0.0046	75.68
Bi-LSTM (± 50%)	0.0026	91.95
Bi-LSTM (± 20%)	0.0025	92.81
Bi-LSTM (Optimized)	**0.0022**	**96.65**

The best performance is highlighted in bold

pronounced, as highlighted in the blue boxes. Nevertheless, the models without prior RMS accelerations were also able to capture overall trends, demonstrating their predictive capability even in the absence of the most significant feature.

In this regard, Fig. 13 compares the RMSE and normalized RMSE (nRMSE) between true and predicted RMS acceleration according to the range of the vibration amplitude. The nRMSE is calculated by dividing the RMSE by the standard deviation of the corresponding true values of the sequentially predicted RMS accelerations. The x-axis represents binned ranges of RMS accelerations, categorized similarly to a histogram.

As anticipated, both RMSE and nRMSE were minimal in regions of relatively low amplitude. The majority of errors were concentrated in the high-amplitude range of RMS acceleration exceeding 0.06 m/s^2, which typically occurs during daytime periods with high traffic volumes. This performance degradation can be attributed to the absence of the key feature, prior RMS acceleration, resulting in reduced predictive accuracy in high-amplitude regions where the available data are relatively sparse. Conversely, in amplitude ranges where the majority of measurement data is concentrated, the model demonstrated relatively reliable predictions despite the absence of the

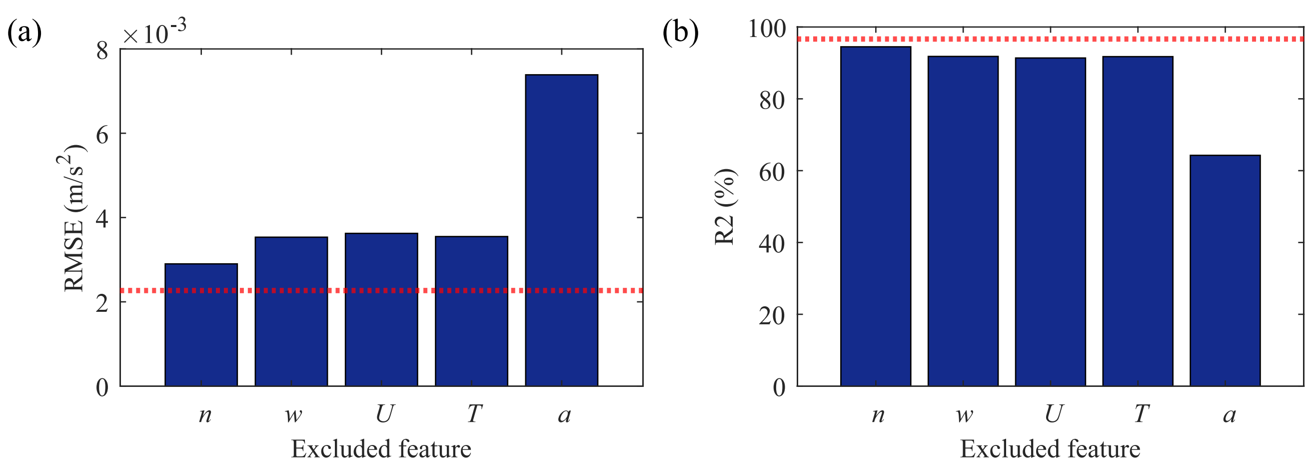


Fig. 11 Initial feature engineering (τ =3, α= 0.3): (**a**) RMSE and (**b**) R2

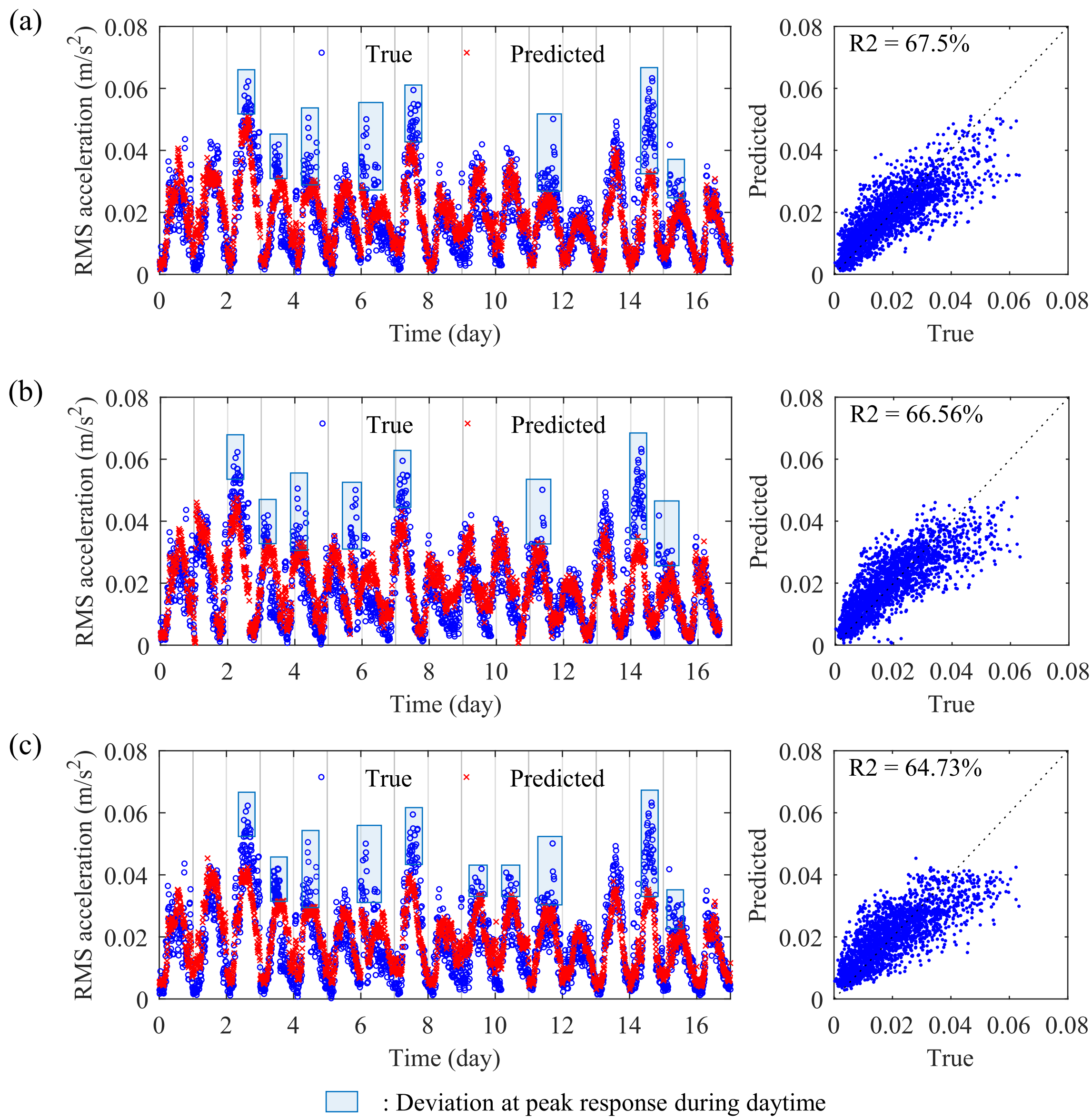


Fig. 12 The results of predicted vibrational response without a prior RMS acceleration ($\alpha = 0.3$): (**a**) $\tau = 1$, (**b**) $\tau = 2$, and (**c**) $\tau = 3$

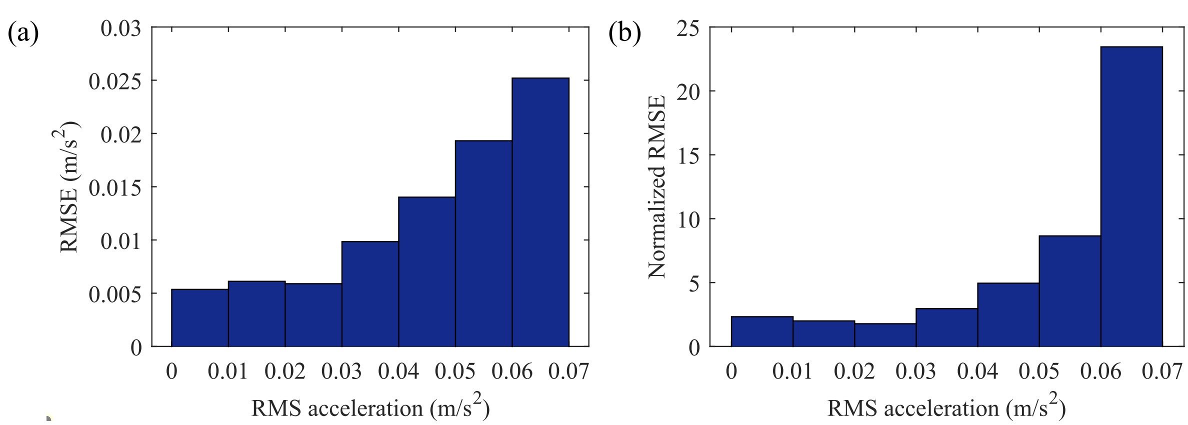


Fig. 13 (**a**) RMSE and (**b**) nRMSE according to the vibration amplitude ranges ($\tau = 1$)

key feature, likely due to the availability of sufficient training data in those regions.

5 Anomaly detection using forecasting models

In addition to forecasting dynamic responses, this section extends the proposed framework to detect anomalies in vibrational behavior. By analyzing the residuals between predicted and measured responses, the framework identifies abnormal events, such as VIVs, that deviate significantly from expected trends. Two anomaly detection models are evaluated: one excluding prior RMS acceleration from the input configuration to enhance sensitivity to abrupt changes, and the other including prior RMS acceleration to improve prediction accuracy for regular conditions. The results are validated using field monitoring datasets, including both normal operational data and VIV events, demonstrating the framework's robustness in detecting hazardous vibrations.

5.1 Anomaly detection

The anomaly detection capability was incorporated by leveraging the residuals between the predicted and measured vibrational responses. Specifically, the model first forecasts the RMS acceleration ($\hat{a}$) using the Bi-LSTM. The residuals, defined as the difference between the measured RMS acceleration (a) and the forecasted RMS acceleration ($\hat{a}$), are then analyzed to identify anomalies as Residual $= |a - \hat{a}|$. The anomaly detection threshold was then defined as the 90% percentile of the residuals calculated during normal operation (D1 in this study).

As each data point represents a 10-min interval, the first anomaly detection involves identifying "point anomalies", which classify a residual value from a single data point exceeding the predefined threshold as an anomaly. The second step focuses on detecting "subsequence anomalies" to ensure robustness against transient fluctuations and to reduce false positives caused by localized prediction errors. In this approach, anomalous vibrations are identified when consecutive residuals exceed the threshold. A subsequence outlier is defined as abnormal behavior persisting for 30 min, which corresponds to three consecutive data points.

The two prediction models were employed for anomaly detections: M1 excluding prior RMS acceleration from the input data and M2 including prior RMS acceleration. Both models were trained with a time lag of 30 min and α of 0.3. The effectiveness of these response prediction models in detecting anomalies was validated using the D1 dataset, which consists of normal operational data, as well as a one-week field monitoring dataset (D2) that includes both normal operations and VIV events.

5.2 Results and discussions

Figure 14 presents the anomaly detection results for M1. Both point and subsequence anomaly detection approach accurately identified the VIV events in the D2 dataset. Even though several false positives were only found from point anomaly detections in the D1 dataset, which represents normal operation, these were resolved by using subsequence anomaly detection approach. As discussed in Sect. 4.4, the prediction performance of M1 is somehow degraded due to the absence of prior RMS acceleration information. However, this limitation is counterbalanced by a higher threshold in the anomaly detection stage, which effectively prevents

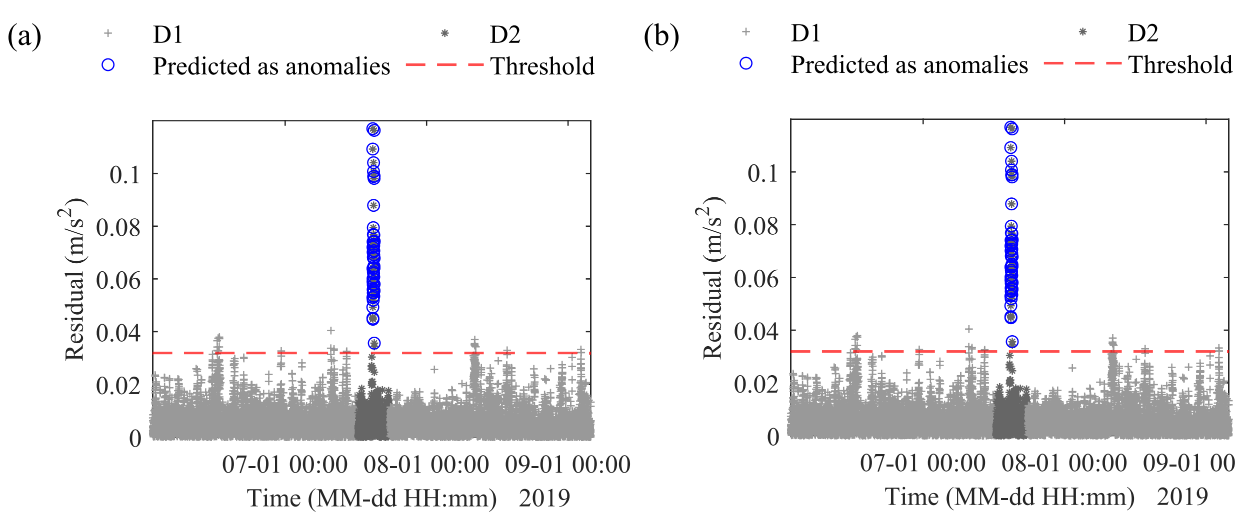


Fig. 14 Anomaly detection results of M1: (**a**) point and (**b**) subsequence anomaly detection

false positive errors. In addition, since M1 does not rely on prior bridge responses, it performs better in detecting sudden anomalies that deviate significantly from expected trends.

Figure 15 illustrates the anomaly detection results for M2. While M2 exhibited superior predictive performance compared to M1, with R2 of 96.65%, its anomaly detection results were less robust. Specifically, for point anomaly detection, M2 accurately identified the VIV events in D2; however, it also resulted in considerable false positives in the D1 dataset. This increased sensitivity can be attributed to the much lower residual values in M2 compared to M1, which, while indicative of better prediction accuracy, also led to a lower threshold that was less robust against localized prediction errors.

Most of these localized false positives were resolved during the subsequence anomaly detection, as shown in Fig. 15(b). However, M2 with the subsequence anomaly detection completely failed to detect the continuous VIV

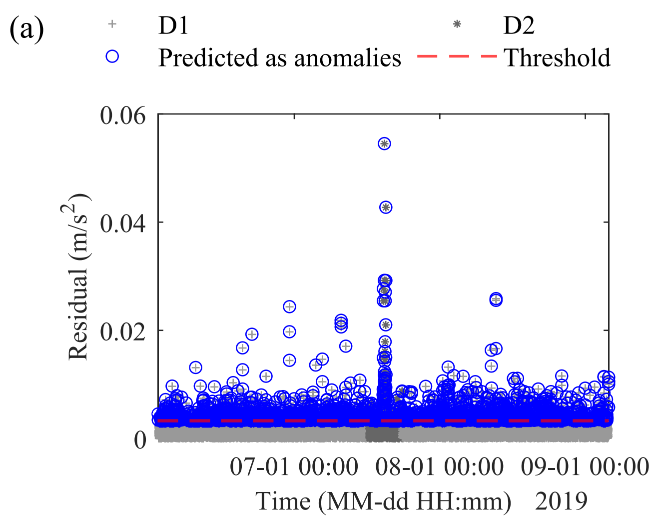


Fig. 15 Anomaly detection results of M2: (**a**) point and (**b**) subsequence anomaly detection

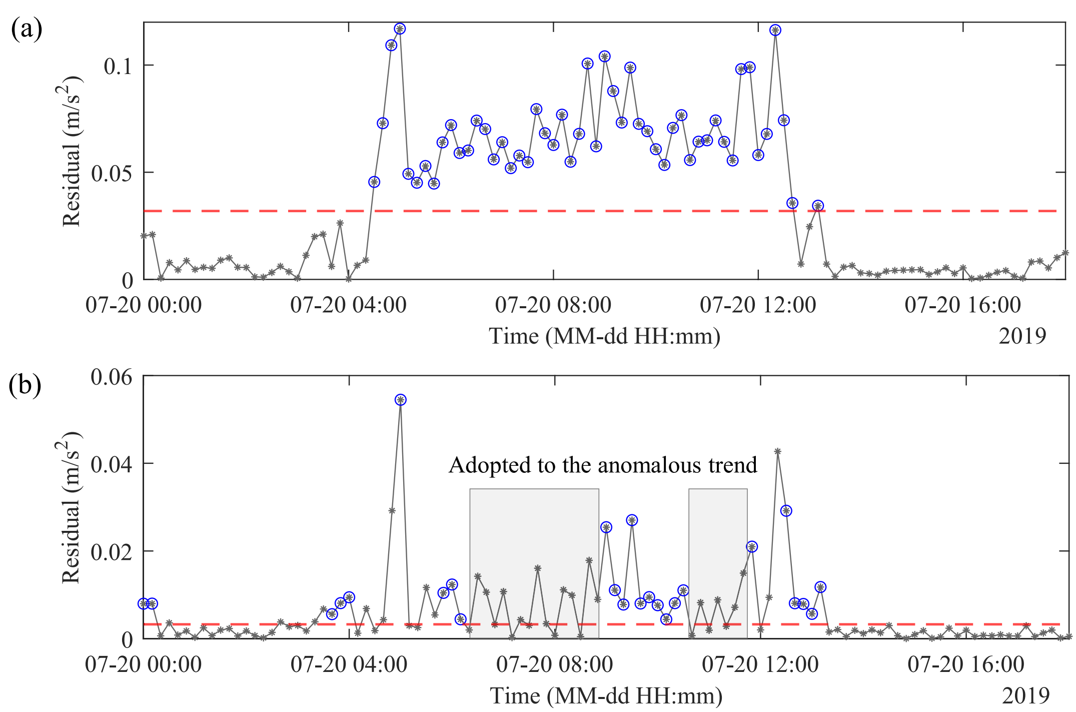


Fig. 16 Zoomed control chart of the subsequence anomaly detection: (**a**) M1 and (**b**) M2

events in D2. This failure can be attributed to the reliance of M2 on prior RMS acceleration information. As shown in Fig. 16(b), the initial VIV cases in D2 exceeded the threshold when M2 was applied. However, as the model incorporated subsequent measurements of the abnormal vibration amplitudes as prior RMS acceleration data, it adapted to these anomalous trends and even began to estimate the anomalous vibration amplitudes accurately. In other words, even though anomalous datasets were not included in the training datasets, the inclusion of prior RMS acceleration led to overfitting, as the model attempted to replicate historical vibration trends closely. Consequently, after the first 2–3 data points, the residuals decreased rapidly, preventing the detection of sustained VIV events.

In contrast, as shown in Fig. 16(a), the independence of M1 from prior RMS trends—achieved by excluding prior RMS acceleration and relying solely on EOCs for amplitude estimation—enabled it to successfully detect continuously occurring anomalous behavior, even when the behavior deviated significantly from previous response patterns. This improvement can be attributed to the reduced dependency of M1 on historical vibration trend, which allowed the model to focus more on EOCs as the primary predictors of anomalous behavior. By excluding prior RMS acceleration, M1 avoided overfitting, a limitation often observed when prior RMS acceleration is included. Consequently, M1 demonstrated greater sensitivity to detecting anomalies, particularly for VIV events, while maintaining a lower false positive rate. These findings underscore the critical importance of input configuration in optimizing the performance of data-driven anomaly detection frameworks, especially in scenarios where reliance on historical responses might obscure ongoing abnormal behavior.

6 Conclusions

This study proposed a data-driven framework for forecasting dynamic responses and detecting anomalies, such as vortex-induced vibrations (VIVs), in long-span bridge decks. The approach utilized a Bayesian-optimized bidirectional long short-term memory (Bi-LSTM) network, employing a sequence-to-sequence regression model. By leveraging real-world SHM and WIM datasets collected from an operational long-span bridge, the framework demonstrated its practical applicability for structural health monitoring and early warning systems. The results highlight several key findings and contributions:

1. The proposed framework achieved high accuracy in forecasting vibrational amplitudes across varying time lags. The model exhibited exceptional performance for short-term predictions (time lag $\leq$ 30 min), with RMSE values below 0.05 and R2 exceeding 0.95. These results validate the framework's effectiveness in capturing complex temporal dependencies between environmental and operational conditions (EOCs) and dynamic responses.
2. The framework effectively identified anomalies by analyzing residuals between predicted and measured vibrational responses. Excluding prior RMS acceleration from the input configuration significantly improved anomaly detection performance with near-perfect results. This superior performance is attributed to the model's ability to focus on EOCs as primary predictors, avoiding overfitting to historical vibration trends.
3. Excluding prior RMS acceleration from the input EOCs proved advantageous for subsequence anomaly detection, such as continuous VIV events. The independence from prior RMS trends allowed the model to focus solely on the other EOCs for amplitude estimation, enabling it to detect anomalies that deviate significantly from previous response patterns. In contrast, configurations relying on prior RMS acceleration showed limitations in detecting such subsequence anomalies due to overfitting, as the model adapted to anomalous trends.
4. The proposed method successfully detected VIV events during high wind conditions, showcasing its potential for real-time monitoring of hazardous vibrations. The integration of anomaly detection within the forecasting framework enables proactive maintenance strategies and supports early warning systems to mitigate risks to bridge safety and serviceability.

Despite these contributions, further investigation is required on the model's performance in detecting rare or subtle anomalies, such as small deviations in vibration patterns. Additionally, the framework was validated on a single bridge, and its generalizability to other bridge types or environmental conditions remains to be explored. Since the training data were limited to summer months (June–September), the model's adaptability to seasonal variations, particularly under winter conditions with different temperature and traffic patterns, needs further evaluation. Therefore, future studies should address these limitations by incorporating hybrid approaches that combine data-driven methods with physical models to improve long-term forecasting and anomaly detection. Expanding the dataset to include multiple bridges across diverse environments and a full annual cycle will further validate the framework's robustness and generalizability.

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Data availability The data are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that there is no conflict of competing financial interests or personal relationships that could have appeared to influence the publication of this paper.

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